

Assignment 1: AI System Specification

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Topic: PEAS Framework and Task Environment Analysis

**Selected System: AI Tutor for Personalized Student Learning
Recommendations**

Description of the AI Tutor System:

An AI Tutor is an intelligent software system that monitors a student's learning activities and provides customized study materials, quizzes, and feedback. It continuously learns from student interactions and adapts its recommendations accordingly.

Task 1: PEAS Specification

Performance Measures:

1. Learning outcome improvement rate (pre-test vs. post-test score gain per student).
2. Student engagement duration (average time-on-task per session).
3. Concept mastery accuracy (percentage of topics marked proficient within a set timeframe).
4. Adaptive recommendation precision (percentage of recommended exercises that match the student's current knowledge gap).
5. Dropout/disengagement rate reduction compared to a non-adaptive baseline.

Environment:

The agent operates within a digital e-learning platform accessible via web browsers or mobile apps. Key external factors include the student's prior knowledge level, learning pace, attention span, available study time, internet

connectivity quality, and the curriculum structure defined by educators. The environment also includes school policies, grade level requirements, and multilingual or accessibility needs.

Actuators:

- Content delivery engine (renders text lessons, videos, and interactive exercises).
- Recommendation engine (selects and sequences next learning modules).
- Feedback generator (provides written hints, explanations, and encouragement messages).
- Progress dashboard updater (updates student and teacher-facing reports).
- Notification system (sends reminders, alerts, and motivational prompts via email or app).

Sensors:

- Response input analyzer (captures student answers, response time, and error patterns).
- Clickstream tracker (monitors navigation behavior, time spent per page, and scroll depth).
- Quiz and assessment parser (reads scores and identifies incorrect answer patterns).
- Student profile reader (accesses historical performance data, learning style tags, and grade level).
- Natural language input processor (interprets open-ended typed questions from students).

Task 2: Environment Classification

Fully Observable vs. Partially Observable:**Choice: Partially Observable**

Justification: The agent cannot directly observe the student's cognitive state, emotional engagement, or offline study habits — it only infers understanding from digital interaction signals.

Single-agent vs. multi-agent:**Choice: Multi-agent**

Justification: The system may interact alongside teacher dashboards, parent monitoring tools, and peer collaboration modules, all of which influence the learning environment simultaneously.

Deterministic vs. Stochastic:**Choice: Stochastic**

Justification: Student responses and learning progress are unpredictable — the same lesson delivered twice may yield different outcomes due to mood, fatigue, or prior context.

Episodic vs. Sequential:**Choice: Sequential**

Justification: Each learning session builds directly on previous ones; a student's current knowledge state depends on the entire history of past interactions, not just the current session.

Static vs. Dynamic:**Choice: Dynamic**

Justification: The environment changes continuously as the student answers questions, skips content, or returns after a break, requiring the agent to update its model in real time.

Discrete vs. Continuous:**Choice: Continuous**

Justification: Student knowledge levels, engagement scores, and recommendation weights are all represented as continuous variables that shift gradually over many interactions.

Task 3: Critical Analysis

Question 1 — Greatest Technical Challenge:

The most technically challenging dimension is Partial Observability. The AI Tutor never has direct access to what a student truly understands — it can only observe behavioral signals such as answer correctness, time taken, and click patterns. A student might guess correctly without understanding, or answer incorrectly due to a typo rather than a knowledge gap. Building an accurate internal model of student knowledge (often called a "student model" or knowledge tracing model) from these noisy, indirect signals is genuinely difficult. Errors in this model cascade into poor recommendations, causing the system to either under-challenge advanced students or overwhelm struggling ones — both outcomes leading to disengagement and reduced learning outcomes.

Question 2 — Utility Function:

A meaningful trade-off in this system is between Recommendation Speed (how quickly the next module is suggested) and Recommendation Accuracy (how well it matches the student's actual knowledge gap).

Proposed utility function:

$$U = 0.7 \times \text{Accuracy} - 0.3 \times \text{Latency}$$

Here, Accuracy measures how precisely the recommended content targets the student's weak areas, and Latency measures the delay before the recommendation is served. The current weights prioritize getting the right recommendation over delivering it instantly.

If the weight on Accuracy were doubled (effectively making it $U = 0.9 \times \text{Accuracy} - 0.1 \times \text{Latency}$), the agent would tolerate longer processing times in order to run deeper knowledge-tracing computations before surfacing a recommendation. In practice, this would mean the system performs more background analysis per session, potentially running heavier machine learning inference before each

module selection. This is beneficial in high-stakes educational contexts (exam preparation) but could frustrate younger students expecting instant feedback.

Task 4: Structured Data Design — Structured Representation:

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{  
  "system_name": "AI Tutor for Personalized Student Learning Recommendations",  
  "peas": {  
    "performance_measures": [  
      "learning_outcome_improvement_rate (pre vs post test score gain)",  
      "student_engagement_duration (avg time-on-task per session in minutes)",  
      "concept_mastery_accuracy (percentage of topics marked proficient)",  
      "adaptive_recommendation_precision (match rate to knowledge gap)",  
      "dropout_rate_reduction (compared to non-adaptive baseline)"  
    ],  
    "environment": "Digital e-learning platform accessed via web or mobile, with  
variable student knowledge levels, learning paces, curriculum constraints, and  
connectivity conditions",  
    "actuators": [  
      "content_delivery_engine",
```

```
"recommendation_engine",
"feedback_generator",
"progress_dashboard_updater",
"notification_system"
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"sensors": [
  "response_input_analyzer (answers, response time, error patterns)",
  "clickstream_tracker (navigation, time-on-page, scroll depth)",
  "quiz_and_assessment_parser (scores, incorrect answer patterns)",
  "student_profile_reader (historical performance, learning style, grade level)",
  "natural_language_input_processor (typed student questions)"
]
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"environment_classification": {
  "observability": {
    "choice": "Partially Observable",
    "justification": "The agent cannot observe cognitive state or offline study
behavior, only inferring understanding from digital interaction signals"
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  "agents": {
    "choice": "Multi-agent",
    "justification": "Operates alongside teacher dashboards, parent monitoring
tools, and peer collaboration modules that influence the shared environment"
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"determinism": {  
  "choice": "Stochastic",  
  "justification": "Student responses are unpredictable as the same lesson may  
yield different outcomes due to mood, fatigue, or changing context"  
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"episodicity": {  
  "choice": "Sequential",  
  "justification": "Each session builds on prior ones; the student's current state  
depends on the full history of past interactions"  
},  
"dynamism": {  
  "choice": "Dynamic",  
  "justification": "The environment changes continuously as the student  
interacts, requiring real-time model updates"  
},  
"continuity": {  
  "choice": "Continuous",  
  "justification": "Knowledge levels, engagement scores, and recommendation  
weights are continuous variables shifting gradually across interactions"  
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},  
"utility_function": "U = 0.7 * recommendation_accuracy - 0.3 *  
recommendation_latency"  
}
```